



BAT54XY

Schottky barrier diode

16 July 2026

Product data sheet

1. General description

Schottky barrier quadruple diode with an integrated guard ring for stress protection. Two electrically isolated dual Schottky barrier diodes series, encapsulated in a very small SOT363-3 (SC-88) Surface-Mounted Device (SMD) plastic package.

2. Features and benefits

- Low forward voltage
- Low capacitance
- AEC-Q101 qualified

3. Applications

- Ultra high-speed switching
- Line termination
- Voltage clamping
- Reverse polarity protection

4. Quick reference data

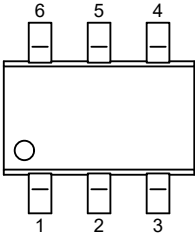
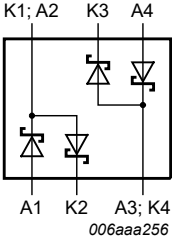
Table 1. Quick reference data

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
Per diode							
V_R	reverse voltage			-	-	30	V
I_F	forward current			-	-	200	mA
V_F	forward voltage	$I_F = 10 \text{ mA}$; $T_{\text{amb}} = 25 \text{ }^\circ\text{C}$	[1]	-	-	400	mV

[1] Pulsed test: $t_p \leq 300 \text{ } \mu\text{s}$; $\delta \leq 0.02$

5. Pinning information

Table 2. Pinning information

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	A1	anode 1	 Transparent top view TSSOP6 (SOT363-3)	 006aaa256
2	K2	cathode 2		
3	A3 / K4	anode3 / cathode4		
4	A4	anode4		
5	K3	cathode3		
6	K1 / A2	cathode1 / anode2		

6. Ordering information

Table 3. Ordering information

Type number	Package		
	Name	Description	Version
BAT54XY	TSSOP6	Plastic surface-mounted package; 6 leads	SOT363-3

7. Marking

Table 4. Marking codes

Type number	Marking code[1]
BAT54XY	%C5

[1] % = placeholder for manufacturing site code

8. Limiting values

Table 5. Limiting values

In accordance with the Absolute Maximum Rating System (IEC60134).

Symbol	Parameter	Conditions	Min	Max	Unit
Per diode					
V _R	reverse voltage		-	30	V
I _F	forward current		-	200	mA
I _{FRM}	repetitive peak forward current	t _p ≤ 1 s; δ ≤ 0.5	-	300	mA
I _{FSM}	non-repetitive peak forward current	t _p < 10 ms; T _j (init) = 25 °C	-	600	mA
P _{tot}	total power dissipation	T _{amb} ≤ 25 °C	[1]	220	mW
T _j	junction temperature		-	125	°C
T _{amb}	ambient temperature		-55	125	°C
T _{stg}	storage temperature		-65	150	°C

[1] Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint.

9. Thermal characteristics

Table 6. Thermal characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
Per diode							
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1]	-	-	460	K/W

[1] Device mounted on an FR4 PCB, single-sided copper, tin-plated and standard footprint.

10. Characteristics

Table 7. Characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
Per diode							
V_F	forward voltage	$I_F = 0.1\text{ mA}$; $T_{amb} = 25\text{ °C}$	[1]	-	-	240	mV
		$I_F = 1\text{ mA}$; $T_{amb} = 25\text{ °C}$		-	-	320	mV
		$I_F = 10\text{ mA}$; $T_{amb} = 25\text{ °C}$	[1]	-	-	400	mV
		$I_F = 30\text{ mA}$; $T_{amb} = 25\text{ °C}$	[1]	-	-	500	mV
		$I_F = 100\text{ mA}$; $T_{amb} = 25\text{ °C}$	[1]	-	-	800	mV
I_R	reverse current	$V_R = 25\text{ V}$; $T_{amb} = 25\text{ °C}$	[1]	-	-	2	µA
C_d	diode capacitance	$V_R = 1\text{ V}$; $f = 1\text{ MHz}$; $T_{amb} = 25\text{ °C}$		-	-	10	pF
t_{rr}	reverse recovery time	When switched from $I_F = 10\text{ mA}$ to $I_R = 10\text{ mA}$; $R_L = 100\text{ }\Omega$; measured at $I_R = 1\text{ mA}$.		-	-	5	ns

[1] Pulsed test: $t_p \leq 300\text{ }\mu\text{s}$; $\delta \leq 0.02$

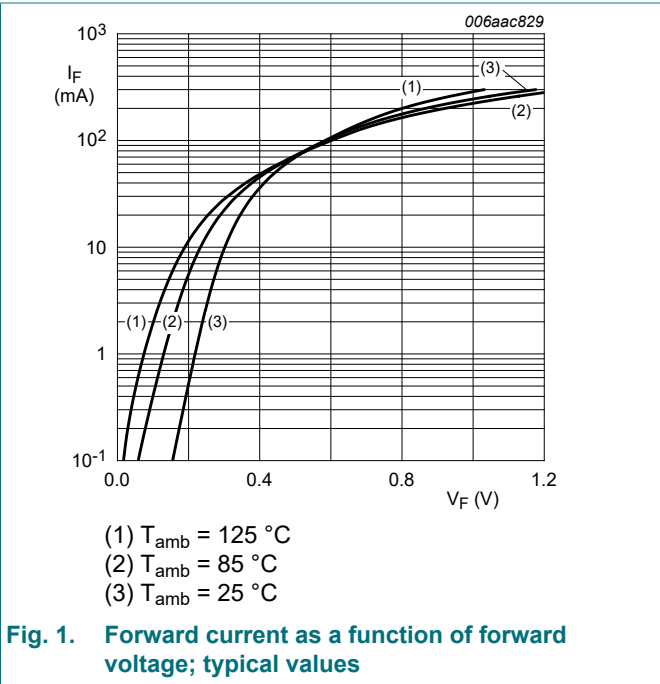


Fig. 1. Forward current as a function of forward voltage; typical values

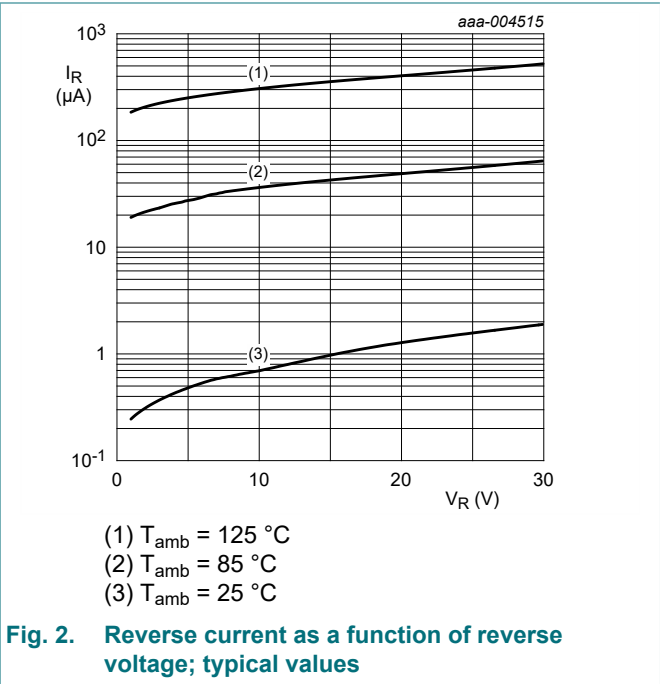
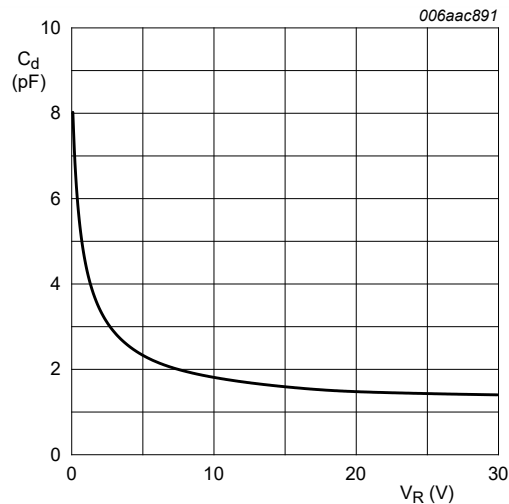


Fig. 2. Reverse current as a function of reverse voltage; typical values



T_{amb} = 25 °C; f = 1 MHz

Fig. 3. Diode capacitance as a function of reverse voltage; typical values

11. Test information

Quality information

This product has been qualified in accordance with the Automotive Electronics Council (AEC) standard Q101 - *Stress test qualification for discrete semiconductors*, and is suitable for use in automotive applications.

12. Package outline

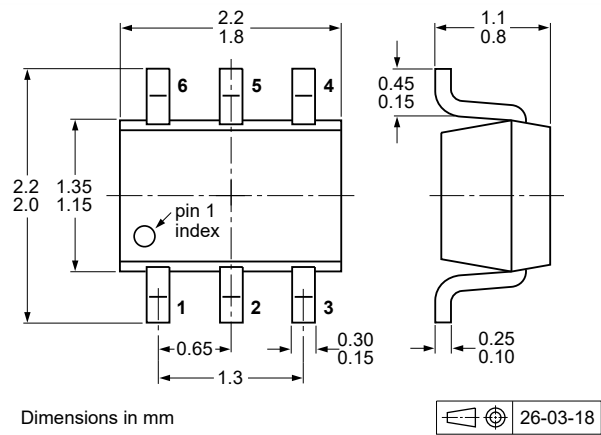


Fig. 4. Package outline TSSOP6 (SOT363-3)

13. Soldering

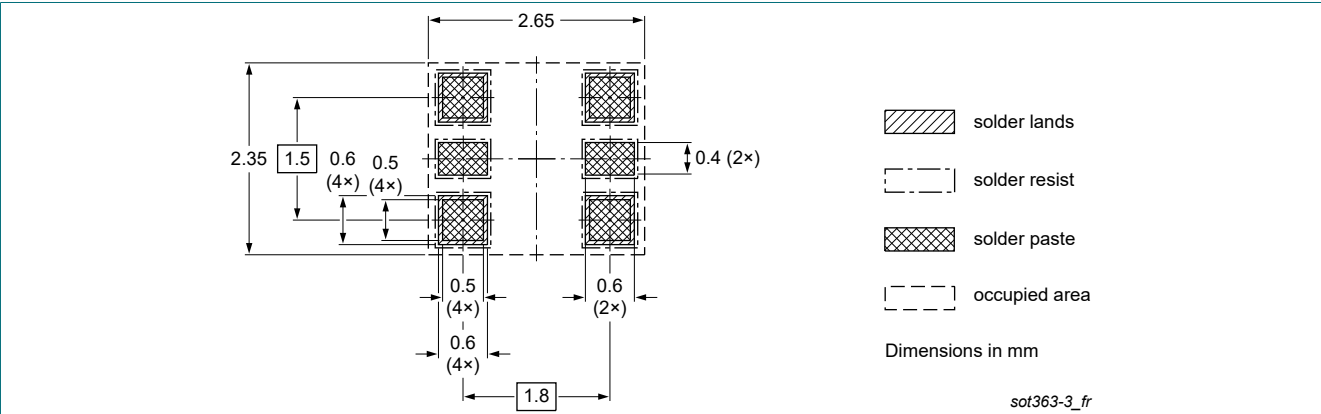


Fig. 5. Reflow soldering footprint for TSSOP6 (SOT363-3)

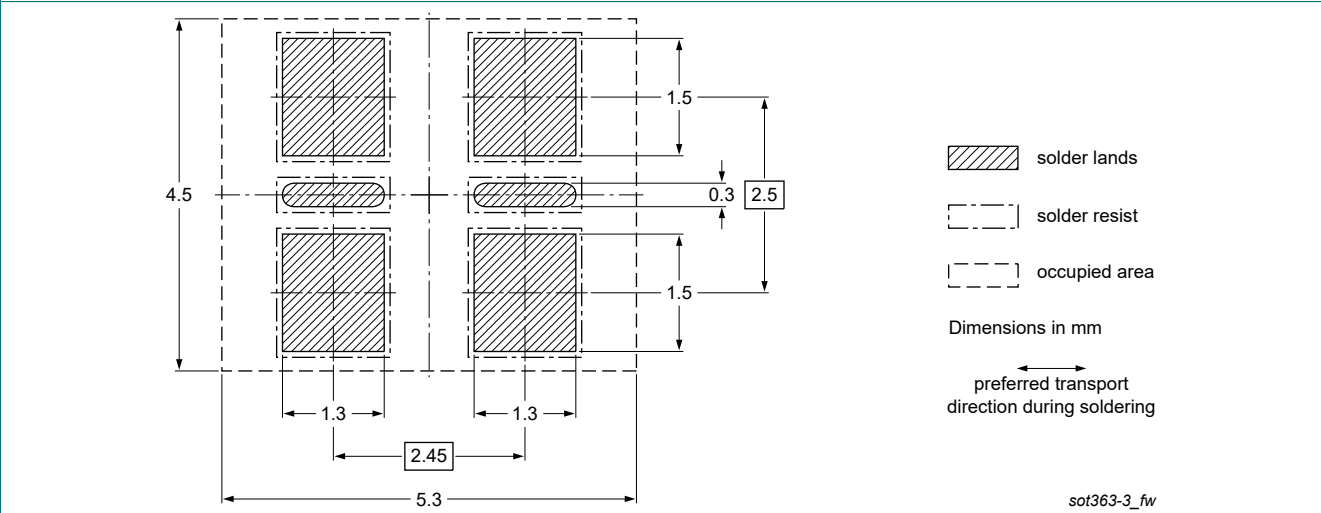


Fig. 6. Wave soldering footprint for TSSOP6 (SOT363-3)

14. Revision history

Table 8. Revision history

Data sheet ID	Release date	Data sheet status	Change notice	Supersedes
BAT54XY v.5	20260716	Product data sheet	-	BAT54XY v.4
Modifications:	• Package changed from SOT363 to SOT363-3			
BAT54XY v.4	20190212	Product data sheet	-	BAT54XY v.3
BAT54XY v.3	20121008	Product data sheet	-	BAT54XY v.2
BAT54XY v.2	20100113	Product data sheet	-	BAT54XY v.1
BAT54XY v.1	20050117	Product data sheet	-	-

15. Legal information

Data sheet status

Document status [1][2]	Product status [3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

- [1] Please consult the most recently issued document before initiating or completing a design.
- [2] The term 'short data sheet' is explained in section "Definitions".
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